Numerical reconstruction of Schrödinger equations with quadratic nonlinearities

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Abstract

We introduce a numerical framework for reconstructing the potential in two-dimensional semilinear elliptic PDEs with power-type nonlinearities from the nonlinear Dirichlet-to-Neumann map. By applying the higher-order linearization method, we compute the Fourier data of the unknown potential and invert it to recover q. Numerical experiments show accurate reconstructions for both smooth and discontinuous test cases.

Introduction

We study the semilinear elliptic model

$$-\Delta u + q\,u^p = 0 \quad \text{in } \Omega, \qquad u\big|_{\partial\Omega} = f, \qquad (1)$$

with $p \in \mathbb{N}_{\geq 2}$ on a bounded domain $\Omega \subset \mathbb{R}^n$ and Dirichlet data f. The objective is to recover the unknown potential $q = q(x)$ from boundary measurements associated with (1). We study this numerically by using the higher order linearization framework of [1, 2]. The idea is to exploit higher derivatives of the nonlinear Dirichlet-to-Neumann (DN) map to obtain auxiliary linearized equations, thereby enabling recovery in settings where the corresponding linear inverse problem is difficult to resolve.

Forward problem

We adopt a Newton-type approach applied to the following weak formulation:

Find $u \in V$ such that

$$F(u)(v) = \int_\Omega \nabla u \cdot \overline{\nabla v}\, dx + \int_\Omega q\,u^p\,\overline{v}\,dx = 0 \quad \forall v \in V_0,$$

where $V = \{u \in H^1(\Omega, \mathbb{C}) \mid u\big|_{\partial\Omega} = f\}$

and $V_0 := H_0^1(\Omega) = \{v \in H^1(\Omega, \mathbb{C}) \mid v|_{\partial\Omega} = 0\}$.

Theorem

Let $F : V \to V_0'$ be the nonlinear operator associated with the problem (1) on a smooth bounded domain Ω, with V and V_0 as defined above.

(i) If u_n is sufficiently close to the exact solution u, then the linearized problem

$$DF(u_n)[\delta u] = -F(u_n) \quad \text{in } V_0'$$

admits a unique solution $\delta u \in V_0$.

(ii) If the initial guess $u_0 \in V$ is sufficiently close to u, the Newton iteration

$$u_{n+1} = u_n + \delta u_n,$$

is well defined and converges to u in V.

The forward problem is solved via Newton method and FEM.

Reconstruction of the potential

We consider the nonlinear boundary-value problem (1) with $p = 2$

$$\begin{cases} -\Delta u + q\,u^2 = 0, & \text{in } \Omega, \\ u = \varepsilon f, & \text{on } \partial\Omega. \end{cases} \qquad (2)$$

Denote

$$v = \left.\frac{\partial u}{\partial \varepsilon}\right|_{\varepsilon=0}, \qquad w = \left.\frac{\partial^2 u}{\partial \varepsilon^2}\right|_{\varepsilon=0}.$$

The differentiation with respect to ε at $\varepsilon = 0$ is justified, since by [1] the solution map of (2) is C^∞ Fréchet differentiable in the neighbourhood of the origin.

Because $u|_{\varepsilon=0} = 0$ is a solution of (2), differentiating once and a second time with respect to ε and evaluating at $\varepsilon = 0$ gives

$$\begin{cases} -\Delta v = 0, & x \in \Omega, \\ v = f, & x \in \partial\Omega. \end{cases} \qquad \begin{cases} -\Delta w + 2\,q\,v^2 = 0, & x \in \Omega, \\ w = 0, & x \in \partial\Omega. \end{cases}$$

Multiplying the equation for w by v_0 a harmonic function and integrating by parts yields

$$\int_{\partial\Omega} f_0\,\partial_\nu w\; ds = 2\int_\Omega q\,v^2\,v_0\; dx, \quad v_0|_{\partial\Omega} = f_0. \qquad (3)$$

Choice of the harmonic functions

Here we choose $v_0 \equiv 1$, and use combinations of harmonic boundary values of $f_\pm = (v_1 \pm v_2)\big|_{\partial\Omega}$, which gives two corresponding solutions u_+ and u_-.

With the choice of Calderón exponentials

$$v_1(x) = \frac{1}{2}e^{(-\zeta - i\xi)\cdot x/2} \quad \text{and} \quad v_2(x) = \frac{1}{2}e^{(\zeta - i\xi)\cdot x/2},$$

we obtain the two measurements (for each ξ):

$$\frac{d^2}{d\varepsilon^2}\int_{\partial\Omega} f_0 \Lambda_q(\varepsilon f_\pm)\, ds\Big|_{\varepsilon=0} = 2\int_\Omega q v_0 (v_1 \pm v_2)^2\, dx$$

their difference produces:

Reconstruction formula

$$\frac{d^2}{d\varepsilon^2}\int_{\partial\Omega} f_0(\Lambda_q(\varepsilon f_+) - \Lambda_q(\varepsilon f_-))\, ds\Big|_{\varepsilon=0} = 2F(q)(\xi)$$

The measurement is modelled by the boundary integral

$$\int_{\partial\Omega} \partial_\nu (u_+ - u_-)_\varepsilon\, ds, \qquad (4)$$

and to account for measurement errors, we add complex Gaussian noise $\eta(\xi)$ to the measurement.

To estimate the second derivative of noisy measurements, we employ a Savitzky–Golay filter (see [4]).

Numerical reconstruction from Fourier data

We approximate the potential q on $\Omega \subset \mathbb{R}^2$ by a piecewise constant function, this leads to the linear system $F = Eq$ that is solved using Tikhonov or TV depending on the regularization of the potential.

Numerical examples

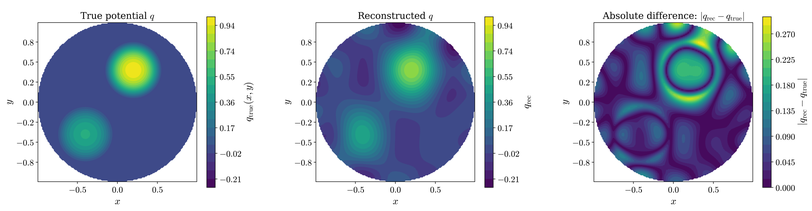


Figure 1:Two smooth bump potential.

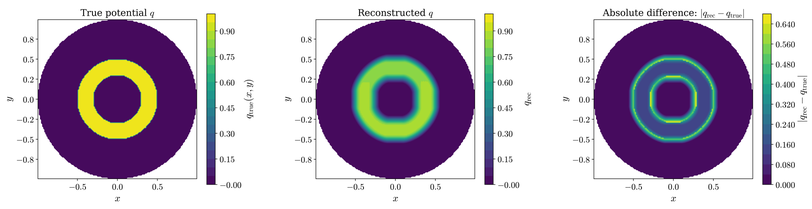


Figure 2:Ring potential.

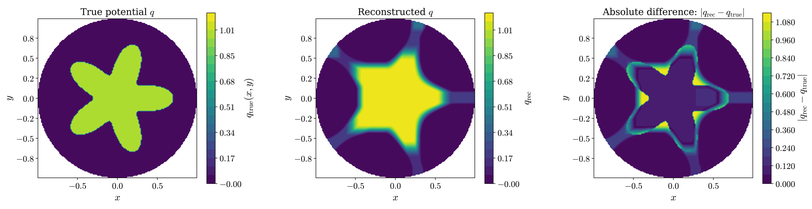


Figure 3:Star shaped potential.

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